

Learning When to Jump for Off-road Navigation

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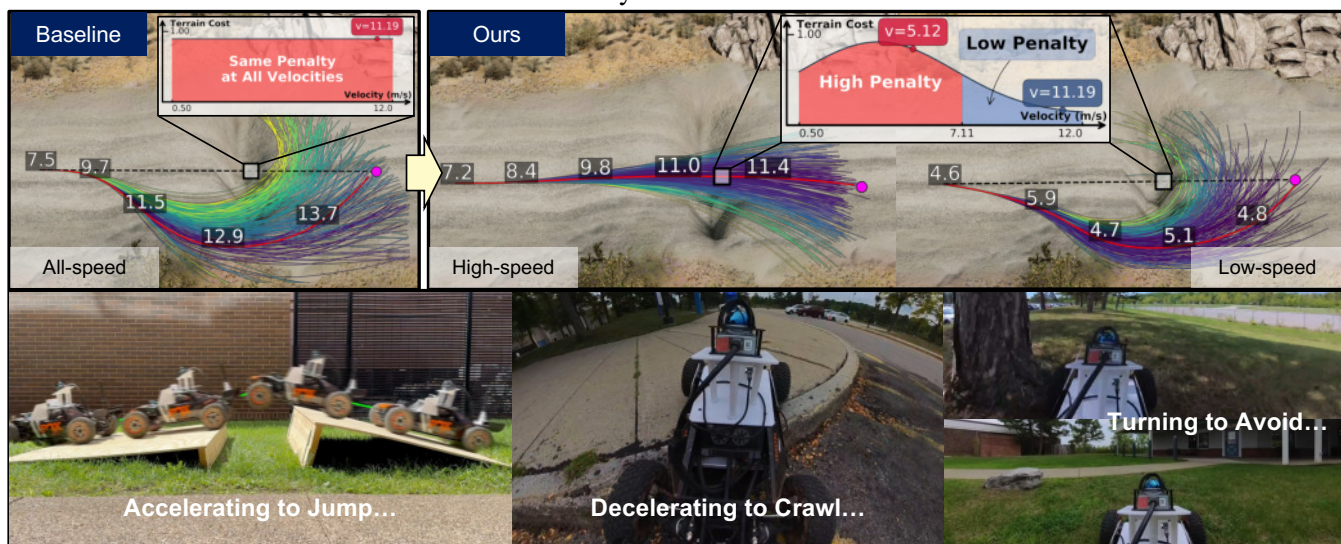


Fig. 1: Challenging terrains often require motion strategies that span from aggressive maneuvers to cautious crawling. Our Motion-Aware Traversability (MAT) model enables the vehicle to decide when to accelerate to jump over a ditch, slow down to crawl over curbs, and steer to avoid obstacles. In contrast, baseline methods assign uniformly high costs to features like ditches, causing the vehicle to avoid them rather than selecting motion strategies that could traverse them more efficiently.

Abstract—Low speed does not always guarantee safety in off-road driving. For instance, crossing a ditch may be risky at a low speed due to the risk of getting stuck, yet safe at a higher speed with a controlled, accelerated jump. Achieving such behavior requires path planning that explicitly models complex motion dynamics, whereas existing methods often neglect this aspect and plan solely based on positions or a fixed velocity. To address this gap, we introduce a Motion-Aware Traversability (MAT) representation to explicitly model terrain cost conditioned on actual robot motion. Instead of assigning a single scalar score for traversability, MAT models each terrain region as a truncated Gaussian function of velocity. During on-line planning, we decompose the terrain cost computation into two stages: (1) predict terrain-dependent Gaussian parameters from perception in a single forward pass, (2) efficiently update terrain costs for new velocities inferred from current dynamics by evaluating these functions without repeated inference. We develop a system that integrates MAT to enable agile off-road navigation and evaluate it in both simulated and real-world environments with various obstacles. Results show that MAT achieves real-time efficiency and enhances the performance of off-road navigation, reducing path detours by 75% while maintaining safety across challenging terrains.

I. INTRODUCTION

Off-road navigation presents unique challenges for autonomous robots [1]–[3], where safety and efficiency depend not only on terrain geometry but also on the robot’s motion dynamics. For example, crossing a ditch at moderate speed may lead to getting stuck at the bottom, but a higher, controlled speed generates momentum to jump over it safely. Conversely, navigating bumpy roads and ruts may only be

safe at low speeds to prevent instability. Such situations highlight that traversability is not a static property of terrain appearance, as it must account for how the robot moves.

Most existing traversability estimation methods overlook this coupling between terrain and motion dynamics. They typically only infer position-dependent cost maps from terrain appearance and assign static scalar values to each location, independent of the robot’s intended motion. While effective for avoiding obstacles or following smooth trails, these methods share a fundamental limitation: they cannot distinguish situations where the same terrain can be safe or unsafe depending on the robot’s speed or motion profile.

To address this gap, we propose a Motion-Aware Traversability (MAT) map that explicitly conditions terrain cost on the robot’s motion dynamics. As illustrated in Fig. 1, each pixel represents a function of velocity rather than a scalar score. The function is a truncated Gaussian over a fixed velocity range, parameterized by terrain-dependent attributes: peak difficulty, the velocity at which it occurs, and sensitivity to deviations. These compact parameters capture richer terrain characteristics than a single score.

We further develop an off-road navigation system by integrating MAT with a Model Predictive Path Integral (MPPI) [4] planner, enabling real-time adaptation of terrain costs to the motion inferred under system dynamics. Computation is two-stage: (1) a single neural inference predicts map-wide Gaussian parameters, and (2) as the motion changes, new terrain costs can be rapidly inferred by calling these functions without re-running the perception network.

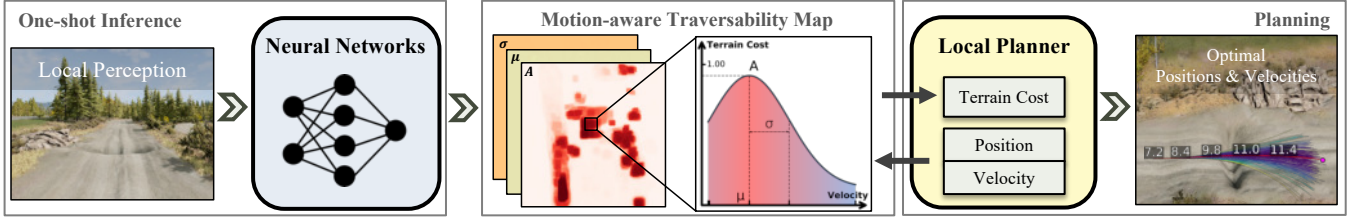


Fig. 2: The architecture of the proposed pipeline. Given a local perception input (a LiDAR-based height map in practice), the neural network performs one-shot inference to generate the Motion-Aware Traversability (MAT) map, where each cell encodes a Gaussian function of velocity. The local planner (e.g., MPPI) then samples trajectories based on vehicle dynamics, continuously querying the MAT map for terrain costs with position-velocity pairs to optimize motion in real time.

II. RELATED WORK

Position-only traversability. Analytic methods derive costs from local slope, roughness, step height, or curvature [5], [6]. Learning-based systems predict geometric or semantic terrain representations [7], [8], use vehicle-terrain interaction signals as self-supervision [9]–[12], or learn from expert trajectories [13]–[15]. These methods improve terrain understanding but normally assign one cost to a location. Consequently, a ditch labeled as hazardous is avoided even when a feasible jump exists, while a mildly rough region may be accepted without expressing the need to slow down.

Motion-enhanced navigation. Physics-based traversability can combine elevation-derived attitude with velocity constraints [16], but quasi-static ground-contact assumptions exclude controlled airborne motion and reduce complex geometry to limited features. HDIF [17] conditions a learned cost on speed, but a changed speed entails another forward pass. Learned speed maps [18], [19] guide a vehicle toward demonstration-like velocities, while a separate motion-independent traversability map still rejects terrain that requires a particular maneuver. MAT complements such work by making the terrain term explicitly dependent on the motion generated inside the planner.

III. APPROACH

A. Motion-Aware Traversability Learning

The core idea of MAT is to predict *terrain-dependent Gaussian parameters* from perception rather than directly regressing the final traversability cost. Our neural model outputs a three-channel map where each position $\mathbf{x}$ is represented by: peak penalty $A(\mathbf{x})$, the velocity at which peak penalty occurs $\mu(\mathbf{x})$, and the sensitivity to velocity deviations $\sigma(\mathbf{x})$. Together, these parameters define a *velocity-conditioned terrain cost function* for each $\mathbf{x}$.

Concretely, given a height map $\mathbf{H} \in \mathbb{R}^{H \times W}$, a network g_θ predicts a three-channel MAT map $\hat{\Psi}$. Each location $\mathbf{x}$ stores parameters $\psi(\mathbf{x}) = [A(\mathbf{x}), \mu(\mathbf{x}), \sigma(\mathbf{x})]^\top$, which define

$$T(\mathbf{x}, v) = A(\mathbf{x}) \exp\left(-\frac{(v - \mu(\mathbf{x}))^2}{2\sigma(\mathbf{x})^2}\right). \quad (1)$$

To learn g_θ , we first construct self-supervised labels Ψ from repeated vehicle-terrain interactions, and then train an image-to-image network with a masked regression objective.

We execute R runs over the same terrain with speeds spanning $[v_{\min}, v_{\max}]$ and resample each trajectory at common

positions $\{\mathbf{p}_i\}_{i=0}^M$ [20], [21]. Given velocity-traversability samples $\mathcal{S}_i = \{(v_{r,i}, t_{r,i})\}_{r=1}^R$, we estimate $\psi(\mathbf{p}_i)$ by fitting Eq. (1). The traversability $t_{r,i}$ jointly measures rollover risk, travel time, and integrated throttle effort. A lightweight U-Net [22] predicts the fitted parameters from the height map.

B. Motion-Aware Navigation System

We develop a navigation system that integrates MAT with perception, planning, and control modules for agile off-road driving. The planning pipeline is initiated by the MAT map and the goal location $\mathbf{g}$ provided by a global waypoint node. Unlike prior systems that optimize only positions to minimize terrain cost, we jointly optimize positions and velocities to produce terrain-adaptive motion strategies.

Each MPPI iteration propagates sampled controls through vehicle dynamics, evaluating trajectories via:

$$\mathbb{J} = \sum_{t=0}^{T-1} [\mathbf{u}_t^\top \mathbf{R} \mathbf{u}_t + T(\mathbf{x}_t, v_t) + c_{\text{aux}}(\mathbf{s}_t, \mathbf{g})] + \phi(\mathbf{s}_T, \mathbf{g}). \quad (2)$$

Because $T(\mathbf{x}_t, v_t)$ is a closed-form function of velocity, it can be rapidly updated for a new velocity sampled during iterative optimization. Given that MAT provides parameters for the entire local environment, the planner can efficiently evaluate any $(\mathbf{x}_t, v_t)$ pairs sampled across all iterations.

IV. EXPERIMENTS

We use a Position-Only Traversability (PO-Trav) that shares the same configuration as MAT but replaces the motion-aware terrain cost with a position-only formulation. For comparison within motion-enhanced traversability, we include AnyNav [23] and PhysORD [24] in simulation. We evaluate navigation using detour distance, traversal time, and energy consumption computed as integrated throttle effort.

A. Simulation: Obstacle Traversal

To evaluate MAT on challenging obstacles, we consider a task in which the vehicle must find a feasible motion plan from a fixed start to a goal across obstacles. We test three terrain geometries (a short ditch, a long ditch, and a bump) at slow, moderate, and fast operating velocities. Under this setup, the only difference between each comparison pair lies in whether the terrain cost is conditioned on velocity.

As summarized in Table I, MAT reduces the detour distance by an average of 75% across all obstacles compared to PO-Trav. MAT also achieves lower traversal time and energy

TABLE I: Quantitative comparison between MAT and the baselines in the obstacle traversal task. For each metric, the average value (**Avg.**) across three velocity levels is reported. In each comparison, the better result is in bold.

Terrain Geometry	Method	Detour Distance (m) ↓				Traversal Time (s) ↓				Energy Consumption ↓			
		T_1	T_2	T_3	Avg.	T_1	T_2	T_3	Avg.	T_1	T_2	T_3	Avg.
Short Ditch	PO-Trav	2.43	2.72	2.54	2.56	18.49	10.89	8.56	12.65	0.78	0.72	0.96	0.82
	PhysORD	2.35	2.75	2.17	2.42	17.96	12.64	8.54	13.05	0.77	0.74	1.00	0.84
	AnyNav	2.60	2.46	2.49	2.52	21.58	12.16	8.65	14.13	0.97	0.81	1.01	0.93
	Ours	2.30	0.06	0.05	0.81	18.25	9.21	8.22	11.90	0.77	0.87	0.78	0.81
Long Ditch	PO-Trav	2.74	3.17	3.77	3.22	16.87	13.66	6.92	12.49	0.91	0.83	1.07	0.94
	PhysORD	2.39	3.27	3.91	3.19	22.66	13.13	7.12	14.30	0.81	0.88	1.09	0.93
	AnyNav	2.51	3.10	4.01	3.21	19.76	12.80	8.49	13.69	0.83	0.79	1.14	0.92
	Ours	0.01	3.62	0.12	1.25	17.07	12.79	5.97	11.94	0.74	0.87	1.06	0.89
Bump	PO-Trav	2.52	2.04	2.94	2.50	17.91	16.29	7.75	13.98	0.87	0.92	1.07	0.95
	PhysORD	2.98	2.17	3.35	2.83	20.15	14.02	7.88	14.02	0.85	0.79	1.05	0.90
	AnyNav	2.49	2.45	2.93	2.62	19.35	14.79	7.29	13.81	0.80	0.84	1.03	0.89
	Ours	0.07	0.06	0.15	0.09	17.45	13.71	8.41	13.19	0.79	0.76	0.98	0.84



Fig. 3: Real-world ditch traversal strategies in manmade (left) and natural (right) environments. While PO-Trav is consistently conservative, MAT demonstrates velocity-dependent reasoning by jumping (Ours) or avoiding (Ours-Slow). Simultaneously, MAT identifies the fence as universally non-traversable. Overlaid trajectories illustrate these distinct motion strategies.

consumption than AnyNav and PhysORD, as their dynamics models remain insufficient to model the coupled effects of terrain and velocity. Overall, MAT produces more adaptive and efficient motion plans than PO-Trav: it avoids the short ditch at low speed and accelerates to jump across it when sufficient speed can be achieved, while it decelerates before crossing a bump that is only traversable at low velocity.

B. Real-world Navigation

We deploy our system on the *Racer* in two ditch traversal experiments: a manmade environment and a natural field. Both scenarios involve obstacles with significant challenges that are typically considered untraversable by prior methods, forcing traditional planners to take long detours. In the manmade setting, the vehicle navigates toward a goal blocked by a ditch and a tall metal fence. As shown in Fig. 3, PO-Trav consistently avoids both obstacles due to the high terrain cost. In contrast, MAT identifies that the ditch is traversable at high speed and accelerates to jump across it, while correctly recognizing the fence as untraversable at all velocities. When using the same MAT model with planner parameters adjusted

to simulate a slow-moving vehicle (“Ours-Slow”), MAT correctly identifies that it cannot reach the velocity required for a safe jump and shifts its strategy to avoid the ditch. In the natural scenario, the vehicle must traverse an uneven, snow-filled ditch. PO-Trav avoids the ditch entirely, whereas MAT exploits higher speeds to traverse it directly and avoids it when operating under low-speed conditions.

V. CONCLUSIONS

This paper introduced Motion-Aware Traversability (MAT), a representation that conditions terrain cost on both terrain perception and vehicle motion for off-road navigation. By modeling traversability as a velocity-dependent function with parameters learned from perception, MAT captures how motion feasibility varies across terrains and enables efficient cost evaluation during planning. Experiments in simulation and the real world show that MAT improves adaptability and efficiency, enabling vehicles to plan agile yet safe maneuvers such as accelerating to jump across ditches, slowing on bumps and curbs, and consistently avoiding obstacles.

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